TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING

ADVANCED COLLEGE OF ENGINEERING AND MANAGEMENT
DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

KALANKI, KATHMANDU

[EX 755]

A MAJOR PROJECT REPORT ON

“Land Use and Land Cover Analysis Using AI with Drone-Captured Aerial Footage”

SUBMITTED BY:

Aaryab Panta	(36701)
Abinawan Bhattarai	(36703)
Nikesh Baral	(36718)
Shubham Raj Paudel	(36729)

SUPERVISED BY:

Er. Amit Kumar Rauniyar

A Major Project Report submitted to the Department of Electronics and Computer Engineering in partial fulfilment of the requirements for Bachelors degree in Electronics, Communication and Information Engineering.

Kathmandu, Nepal

March, 2025

Land Use and Land Cover Analysis Using AI with Drone-Captured Aerial Footage

Submitted By:

Aaryab Panta	(36701)
Abinawan Bhattarai	(36703)
Nikesh Baral	(36718)
Shubham Raj Paudel	(36729)

Supervised By:

Er. Amit Kumar Rauniyar

A MAJOR PROJECT SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR THE DEGREE OF BACHELOR IN ELECTRONICS, COMMUNICATION AND INFORMATION ENGINEERING

Submitted to:

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

ADVANCED COLLEGE OF ENGINEERING AND MANAGEMENT

Kalanki, Kathmandu

March, 2025

LETTER OF APPROVAL

The undersigned certify that they have read and recommended to the Institute of Engineering for acceptance, a project report entitled **"Land Use and Land Cover Analysis Using AI with Drone-Captured Aerial Footage"** submitted by:

Aaryab Panta	(36701)
Abinawan Bhattarai	(36703)
Nikesh Baral	(36718)
Shubham Raj Paudel	(36729)

In the partial fulfillment of the requirements for the degree of Bachelor in Electronics, Communication and Information Engineering.

....

Project Supervisor

Er. Amit Kumar Rauniyar

....

Head of Department

Senior Asst. Prof. Prem Chandra Roy

Department of Electronics and Computer Engineering

....

External Examiner

March, 2025

COPYRIGHT

Head of Department
Department of Electronics and Computer Engineering
Advanced College of Engineering and Management
Kalanki, Kathmandu
Nepal

ACKNOWLEDGEMENT

We take this opportunity to express our deepest and sincere gratitude to our Project coordinator and also our Project supervisor **Er. Amit Kumar Rauniyar**, Department of Electronics and Computer Engineering for his motivating suggestions, invaluable guidance, help and support that have been instrumental and insightful in shaping the proposal and ensuring its technical feasibility.

We express our deep gratitude to **Er. Prem Chandra Roy**, Head of Department of Electronics and Computer Engineering, **Er. Dhiraj Pyakurel**, Deputy Head, Department Of Electronics and Computer Engineering for their regular support, co-operation, and coordination. We also like to express our sincere gratitude to **Er. Ajaya Shrestha** for his advice, help and support, invaluable guidance in this project selection.

We are grateful to Airlift Technology for providing us with the opportunity to learn about drone mapping and helping us use their equipment resources. Finally, the in-time facilities provided by the department and RIU (Research and Innovation Unit) are also equally acknowledgeable. We are also grateful to all our classmates for their help, encouragement and invaluable suggestions.

Aaryab Panta	(36701)
Abinawan Bhattarai	(36703)
Nikesh Baral	(36718)
Shubham Raj Paudel	(36729)

ABSTRACT

Land Use and Land Cover (LULC) classification techniques and methods are the basis for the environmental monitoring, urban planning, and resources management. This project involves developing a web-based interactive platform for conducting land use and land cover analysis using aerial imagery of the Harisiddhi area, which we personally captured with a drone. The functions of the system include geospatial mapping, deep learning segmentation, and interactive visualization to present in detail results of land classification. The frontend uses Leaflet.js to place high-resolution GeoTIFFs over an interactive map to allow the user to select areas for analysis. The selected area is then sent to the FastAPI-enabled backend, where the original image is clipped and deep learning-based segmentation based on the U-net architecture is performed. The U-net model, which was trained on annotated aerial images, is optimized for pixel-wise land cover classification. The model takes an aerial image as input and performs a series of convolutional feature extractions, downsampling, and upsampling while outputting a segmentation mask with five possible land categories: Bareland, Buildings, Roads, Vegetation, and Water. Results are presented live and consist of both the segmentation mask and statistic analysis that shows the situated distribution of land use types. In addition, the system allows users to export data in a way that allows them to download processed results, such as images and CSV reports for further analysis. The entire flow is optimized such that it works efficiently, smoothly, and seamlessly even on devices with limited computing resources. This project represents a successful road toward the automated land classification application of AI and geospatial technologies toward urban development, agriculture, and environment studies.

Keywords: *Aerial Image*, *Deep Learning Segmentation*, *Drone*, *Geospatial Mapping*, *Land Classification*, *Land Use and Land Cover (LULC)*, *U-Net Model*

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

AI	:	Artificial Intelligence
ANN	:	Artificial Neural Network
COCO	:	Common Objects in Context
CNN	:	Convolutional Neural Network
DEM	:	Digital Elevation Model
DL	:	Deep Learning
FURIA	:	Fuzzy Unordered Rule Induction Algorithm
GIS	:	Geographic Information System
GPS	:	Global Positioning System
GTB	:	Gradient Tree Boosting
IoU	:	Intersection over Union
JSON	:	JavaScript Object Notation
LULC	:	Land Use and Land Cover
ML	:	Machine Learning
MLP-ANN	:	Multi-Layer Perceptron Artificial Neural Network
OBIA	:	Object-Based Image Analysis
RF	:	Random Forest
SVM	:	Support Vector Machine
TIFF	:	Tagged Image File Format

LIST OF UNITS AND CONVERSION

ha	:	hectare
m	:	meter
m/s	:	meter per second

CHAPTER 1

INTRODUCTION

Land Use and Land Cover analysis using AI is modern method of surveying particular field of areas without the need of manual work. It will help to identify and map the open fields, buildings, vegetation, and water sources from the footage that is captured by the Drones. Traditionally, doing such task involved a lot of paperwork and tasks like field surveys, hand mapping, and this could consume a lot of time and most of the times was associated with errors. But by the uses of Artificial Intelligence and Machine Learning, this sector has received a significant boost, since it is now more anatomic and precise and consumes less time to analyze large geographical terrains.

1.1 Background

Land Use and Land Cover are important instruments in environmental analysis, as well as for designing urban and country life. Historically, methods such as manual surveying, aerial photo shoot, and satellite imagery analysis have been employed to gather Land Use and Land Cover information. There are other conventional techniques being applied in the analysis of complex systems, which are effective but their major demerits are that they are costly, they take quite a lot of time and are prone to human errors.

Recent advancements in AI and ML have opened new avenues for Land Use and Land Cover analysis. It is worth noting that other models based on artificial intelligence can perform a large number of calculations within a short time and with high accuracy, to define complex characteristics or make prognosis, which formally were impossible earlier. This shift towards automation not only enhances the efficiency and accuracy of Land Use and Land Cover analysis but also enables continuous monitoring and updating of land cover information.

1.2 Motivation

The motivation for this project arises from the increasing concern in the effective land management and planning particularly in the context of urbanization. The land use and land cover data is required for analyzing the challenging issues of the environment such as deforestation, the expansion of cities and management of water resources to be accurate. Thus, by automating this process of land analysis, we can get the timely and accurate data that may be helpful to make the right decisions and to form the sound policies. Additionally, it is essential to show the public that with the help of AI, it is possible to bring changes to traditional industries and come up with solutions to existing challenges within a shorter time.

1.3 Problem Statement

The conventional approach of Land Use and Land Cover classification is comprehensive, time-consuming, and lacks the exactness, which is crucial when making critical decisions. Depending on the case being followed, it is necessary to find a solution that is much faster in the classification and mapping of land use/cover. This project seeks to address these challenges by implementing an AI-driven solution that identifies and classifies the captured footage data to provide precise and accurate Land Use and Land Cover information.

1.4 Objectives

This project's main goal aims to enroll an accurate, error free and less-time consuming analysis of land and its classification using drone captured data regarding land use and land cover.

1.5 Scope and Limitations

This project proposal is to develop an artificial intelligence based software for Land Use and Land Cover mapping and evaluation of the results. It involves data acquisition through aerial platforms to train the machine learning models and then the actual use of the system in the field to create a level of reliability and efficiency.

However, there are certain degrees of limitations that needs to be taken into consideration. Therefore, accuracy of pattern refers to its ability to perfect itself and this is influenced both by the quality and quantity of data fed to the model. Picture definition and climatic conditions should be addressed when it comes to the way a carper performs. Further, the project may find it difficult to probable computational resources and how they might assimilate various datasets. However, the following limitations have been observed; The project aims to present a comprehensive analysis of Land Use and Land Cover data through the use of artificial intelligence.

CHAPTER 2

LITERATURE REVIEW

Land Use and Land Cover (LULC) analysis plays a crucial role for environmental planning and ecological management along with sustainable development. The use of Artificial Intelligence (AI) in (LULC) classification will improve the accuracy and efficiency of this process. Kalantar *et al.* [1] used the combination of two techniques named FURIA and Object-Based Image Analysis (OBIA), to achieve high accuracy in land cover classification using images captured by drone. Their method outperformed other popular and common methods like Decision tree and Support Vector Machine (SVM). Their method achieved accuracy of almost 91% on the test dataset. This showed FURIA's optimized fuzzy rules are effective for accurate land cover extraction from drone images.

Ouma *et al.* [2] studied and worked on different machine learning techniques for mapping complex urban landscapes using the data from the satellite images. They compared and evaluated various ML algorithms used for mapping. Their result showed Random Forest (RF) outperformed other classifiers such as Multi-Layer Perceptron Artificial Neural Network (MLP-ANN), SVM, and Gradient Tree Boosting (GTB) with the accuracy of 92.8%. Their study showed the importance of selecting the best classifier for urban land use and land cover (LULC). Their study recommended combining multiple classifier to increase accuracy. A similar kind of approach was used by Alshari *et al.* [3] where they implemented a effective method of combining ANN and RF algorithms called ANN-RF. Their approach combined two ML algorithms which eventually helped in achieving high accuracy in classifying land cover types from satellite data. In this case also merging these algorithms provided better accuracy. Alem and Sharma [4] applied deep learning methods to face the challenges of LULC classification in remote sensing using satellite image. LULC using satellite image posed challenges like big data and varying image qualities. Their study highlighted a potential of DL methods, most specifically Convolutional Neural Network (CNN) which could help in overcoming these challenges. Similar approach was used by Carranza-García *et al.* [5] where they also used deep learning approach using CNN which proved to achieve greater accuracy compared to traditional methods such as SVM and RF. They showed CNNs are powerful tool for land cover classification, even for diverse type of images. In their study CNN model proved to be the fastest for both training and testing.

Jagannathan *et al.* [6] proposed a hybrid deep learning approach for LULC classification. They proposed the HEVGG19 method, which combines VGG19 with a hot encoding process, utilizing pre-trained ResNet50 data for transfer learning. The proposed model was tested on satellite and aerial images, achieving an accuracy of 98.5% for LULC classification. Their study showed the model's ability to classify and predict land cover changes effectively, showcasing its potential for urban planning and ecological management.

Year	Authors	Published in	Objective	Contribution	Data	Methods	Key Findings
2017	B. Kalantar, S. B. Mansor, M. I. Sameen, B. Pradhan, and H. Z. Shafri	International Journal of Remote Sensing	Improve accuracy in LULC classification using drone images	Images captured by drone	FURIA and Object-Based Image Analysis (OBIA)	Combined FURIA and OBIA for high accuracy	Achieved 91% accuracy, outperforming Decision tree and SVM
2019	M. Carranza-García, J. García-Gutiérrez, and J. C. Riquelme	Remote Sensing	Improve accuracy and efficiency of LULC classification	Used CNNs for land cover classification, showing greater accuracy compared to traditional methods	Radar images from the JPL AirSAR	CNN, compared to SVM and RF	CNNs proved to be the fastest for both training and testing, achieving greater accuracy
2020	A. Alem, S. Kumar	IEEE Conf. Proc.	Address challenges of LULC classification using satellite images	Highlighted the potential of deep learning methods, specifically CNN, for LULC classification	Landsat -4 and Landsat -6 dataset	CNN	CNNs can overcome big data and varying image quality challenges, proving effective for LULC

Year	Authors	Published in	Objective	Contribution	Data	Methods	Key Findings
2021	J. Jagannathan, C. Divya	Ecological Informatics	Improve accuracy in LULC classification using deep learning	Introduced HEVGG19 method with transfer learning	Satellite and aerial images dataset	VGG19 with data trained in ResNet50	Achieved 98.5% accuracy, showcasing the potential for urban planning
2023	Y. O. Ouma, A. Keitsile, B. Nkwae, P. Odirile, D. Moalafhi, J. Qi	European Journal of Remote Sensing	Compare and evaluate ML algorithms for urban LULC mapping	Showed Random Forest (RF) outperformed other classifiers for mapping complex urban landscapes	Land--sat data (1984-2020) Bots--wana	Various ML algorithms: RF, MLP-ANN, SVM, GTB	RF achieved 92.8% accuracy, recommended combining multiple classifiers for better accuracy
2023	E. A. Alshari, M. B. Abdulkareem, and B. W. Gawali	Frontiers in Artificial Intelligence	Combine ML algorithms to improve LULC classification accuracy	Implemented ANN-RF method to achieve high accuracy in classifying land cover types	Land--sat-8 satellite dataset	Combined ANN and RF algorithms	Improved accuracy through merging ANN and RF algorithms

Table 2.1: Literature Review Summary

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Hardware Requirements

- **Drone:** DJI Mavic 3 Enterprise
- **GPU:** Minimum 8GB GDDR6 VRAM
- **CPU:** Multicore (Minimum 8 cores)
- **RAM:** 32GB DDR5

3.1.1 DJI Mavic 3 Enterprise

The DJI Mavic 3 Enterprise drone is an ideal solution for Land Use and Land Cover (LULC) mapping and surveying projects. Equipped with a high-resolution camera and RTK module, it delivers precise data collection, making it a reliable and efficient tool for professionals in the field.[7]

Figure 3.1: DJI Mavic 3 Enterprise

(Source:[online]Available: https://enterprise.dji.com/mavic-3-enterprise/specs)

Specifications:

- **Camera System:** 20MP wide-angle camera with a 4/3-inch CMOS sensor and a 12MP telephoto camera with hybrid zoom.
- **Mapping and Surveying:** RTK module support for high-precision georeferencing, ensuring centimeter-level accuracy.
- **Flight Time:** Up to 45 minutes, allowing for extensive aerial coverage in a single flight.
- **Obstacle Avoidance:** Omnidirectional sensing with advanced collision avoidance features for safe operation.
- **Transmission:** O3 Enterprise transmission system with a range of up to 15 km.
- **Payload Compatibility:** Supports speaker and RTK module for added functionality.

3.2 Software Requirements

- **Flight Control:** DJI Pilot
- **Photogrammetric Processing:** Agisoft Metashape
- **Programming Language:** Python, Javascript
- **IDE/Text Editors:** VScode, Jupyter Notebook
- **Framework:** TensorFlow, FastAPI

3.2.1 DJI Pilot

DJI Pilot is a comprehensive flight control application designed for managing DJI enterprise drones, offering a suite of tools for flight planning, monitoring, and data management. The app provides real-time telemetry and flight data, including altitude, speed, battery status, and GPS information, which enhances situational awareness during flight operations. It supports various intelligent flight modes, such as Waypoints, Follow Me, and Point of Interest, facilitating complex flight missions tailored to specific project requirements. DJI Pilot also allows users to create detailed mission plans, including setting waypoints and adjusting flight parameters, while integrating payload options for specialized tasks. The user-friendly interface simplifies flight control and mission management, making it accessible for both novice and experienced pilots.

3.2.2 Agisoft Metashape

Agisoft Metashape is a photogrammetry software used for processing aerial and satellite imagery to generate high-resolution 3D models and geospatial data.

- **Functionality:** Converts aerial images into detailed 3D models, orthomosaics, and digital elevation models (DEM).
- **Workflow:** Processes drone-captured images using Structure from Motion (SfM) and Multi-View Stereo (MVS) algorithms to create georeferenced maps.
- **Accuracy:** Supports ground control points (GCPs) and RTK/PPK data for high-precision mapping.
- **Outputs:** Generates orthophotos, DEMs, and 3D point clouds for GIS and remote sensing applications.
- **Use Case in Project:** Used for stitching drone-captured images, generating georeferenced maps, and creating digital surface models for land cover analysis.

3.2.3 Label Studio

Label Studio is an open-source data annotation tool designed for labeling and annotating various types of data, including images, text, and videos. It supports multiple annotation formats, enabling tasks such as object detection, segmentation, classification, and keypoint detection. The platform features a customizable interface that allows users to define their own labeling configurations and workflows tailored to specific project needs. Additionally, Label Studio facilitates the export of labeled data in various formats (e.g., JSON, COCO, YOLO) for seamless integration into machine learning pipelines. With its collaboration features, it supports multi-user projects, making it ideal for team-based annotation efforts.

3.2.4 FastAPI

FastAPI is a modern, high-performance web framework for building APIs with Python 3.6+ that leverages standard Python type hints. Known for its speed, FastAPI is one of the fastest frameworks available, thanks to its asynchronous capabilities and reliance on Starlette and Pydantic. The framework automatically generates interactive API documentation using Swagger UI and ReDoc, making it easier for developers to understand and test the API. It emphasizes type safety by utilizing Python type hints to ensure validation, which improves code quality and reduces runtime errors. Additionally, FastAPI supports asynchronous programming, allowing it to handle many requests simultaneously, making it ideal for high-performance applications. Its built-in dependency injection system simplifies the management and reuse of components.

3.3 Functional Requirements

- **User Interaction**: Enable user to select required area from an interactive map and system should match the selected area against the actual geographic location.
- **Data Processing**: System should be able to extract and preprocess the region selected by the user from the geospatial dataset.
- **Land Cover Classification**: Using a trained deep learning model, system should classify the selected area into predefined land cover subclass (i.e. bareland, building, road, vegetation, and water) generating a mask for pixel-wise segmentation.
- **Result Visualization**: System should be able to display the selected region of interest against the segmentation map along with a color-coded legend and summarize the land cover portions along with real area calculation in pie chart or table format.
- **Result Export**: System should be able to store the results in a ZIP archive so that user can download the image of selected region along with its corresponding segment masks and area calculation data locally on their disk.

3.4 Non-Functional Requirements

- **Performance**: The system should be capable of processing and analyzing aerial imagery within a reasonable timeframe, ensuring low latency for user requests.
- **Scalability**: The system should be scalable to handle increased user load and data volume without compromising performance.

- **Reliability**: The system should ensure high availability and reliability with minimal downtime to guarantee continued operation during both data collection and analysis.
- **Usability**: The user interface must be intuitive and user-friendly, allowing users of varying technical expertise to navigate and interact with the application easily.
- **Compatibility**: The system should be compatible with various devices and browsers to ensure accessibility for all users.

CHAPTER 4

SYSTEM DESIGN AND ARCHITECTURE

4.1 Basic System Workflow

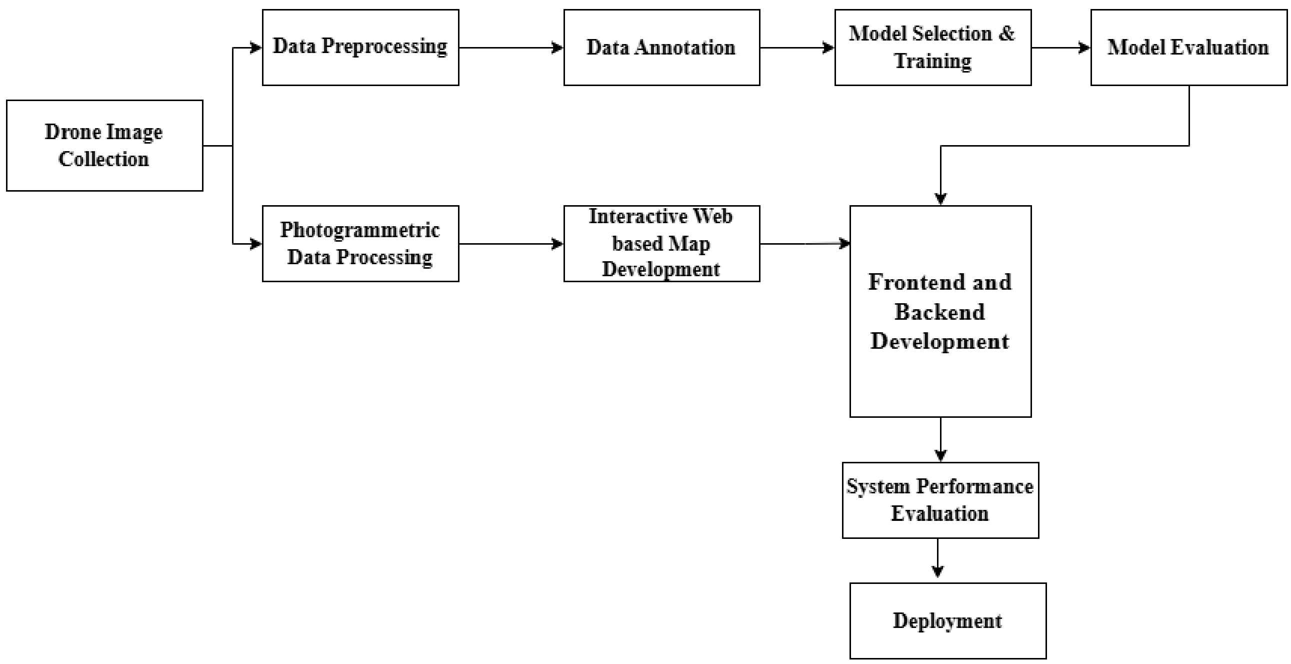

Figure 4.1: Overall System Workflow

- **Drone Image Collection:** The initial stage of the system involves collecting drone images over the designated study area. The drone is deployed to capture high-resolution aerial imagery.

- **Photogrammetric Data Processing:** Photogrammetric data processing is performed on the collected drone images to generate orthomosaic map. This outputs helps in visualizing the spatial structure of the target area and provides useful geospatial information.

- **Data Preprocessing:** Drone image data are processed to enhance quality. This involves filtering, noise reduction, normalization and contrast adjustment.

- **Data Annotation:** This step involves creating ground truth masks through manual annotation. Each image is carefully labeled to identify various land cover classes such as vegetation, water bodies, buildings, roads, and bareland. This annotated dataset serves as the reference standard for supervised learning during model training.

- **Model Selection and Training:** The U-Net deep learning model is selected due to its effectiveness in semantic segmentation tasks. The preprocessed images and their corresponding annotated masks are used to train the model. Data augmentation is done to improve the model's generalization.

- **Model Evaluation:** Once the model is trained, its performance is evaluated using various metrics such as Intersection over Union (IoU) and accuracy. The evaluation process involves comparing the model's predictions with the ground truth annotations.

- **Interactive Web-Based Map Development:** An interactive web-based map is developed to allow users to pan, zoom and select an area of the overlayed GeoTIFF on the map.

- **Frontend and Backend Development:** The web application is built using appropriate frontend and backend technologies. The frontend interface allows users to select area within the GeoTIFF image, view segmentation results, and generate reports. The backend manages the image processing pipeline, model inference, and data storage.

- **System Performance Evaluation:** The overall performance of the system is evaluated based on criteria such as processing speed, accuracy, and model prediction results.

- **Deployment:** The final system is deployed as a web-based application, making it accessible to users to select the desired area on GeoTIFF image and visualize it's output.

4.2 System Architecture

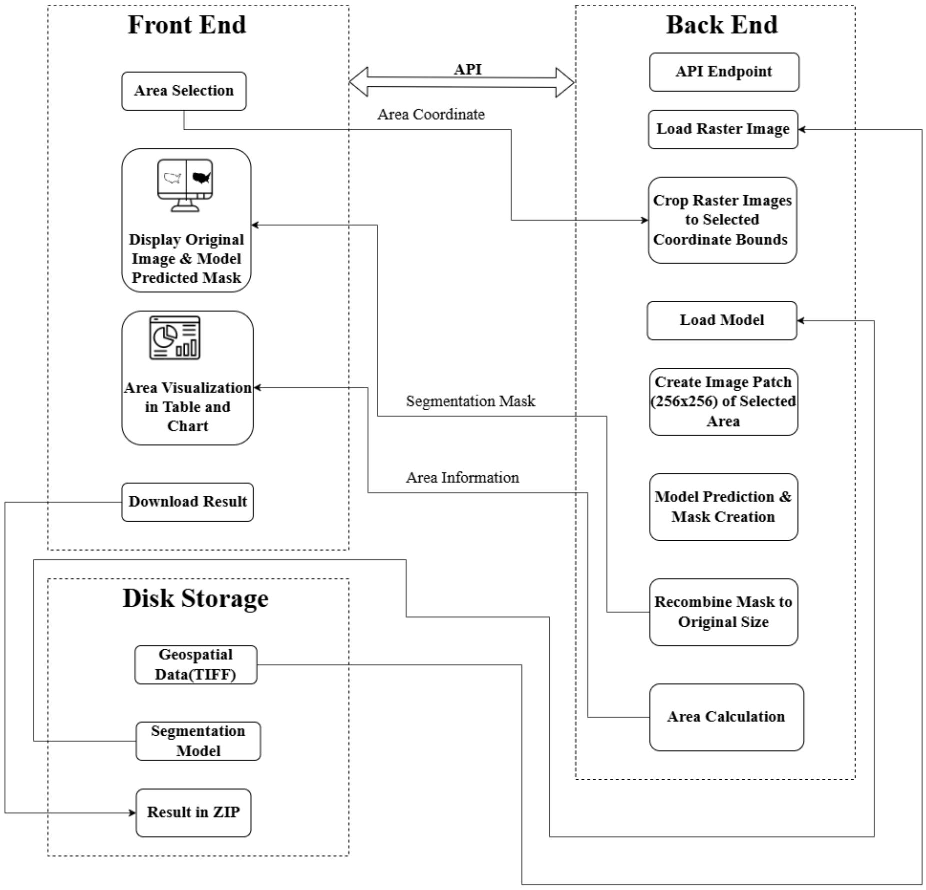

Figure 4.2: System Architecture of LULC analysis

The system architecture for LULC analysis consists of three main components: Frontend, Backend and Disk Storage.

In the Frontend, user are allowed to select an area of interest through interactive interface. The selected area coordinates are sent to the backend via FastAPI endpoint, where the GeoTIFF image is loaded and cropped according to the user selected region. The cropped image is processed and divided into smaller patches of (256x256) pixels which are then passed through deep learning segmentation model U-net to generate the segmentation mask. The mask is recombined into the original image size and area statistics are calculated. The predicted mask and its analysis results are visualized in table and chart in the frontend. User are also provided with the option to download the final results in ZIP format. The disk storage manages Geospatial data, segmentation model, and final downloadable results.

4.3 Sequence Diagram

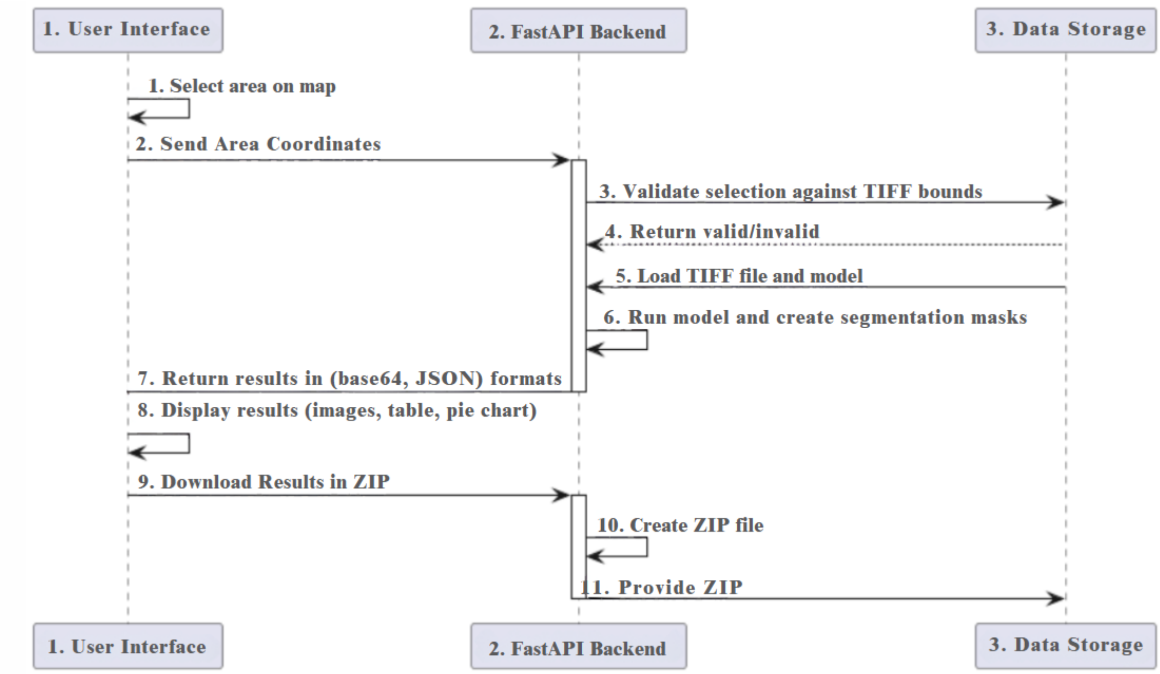

Figure 4.3: Sequence Diagram of LULC analysis system

4.4 U-net Architecture

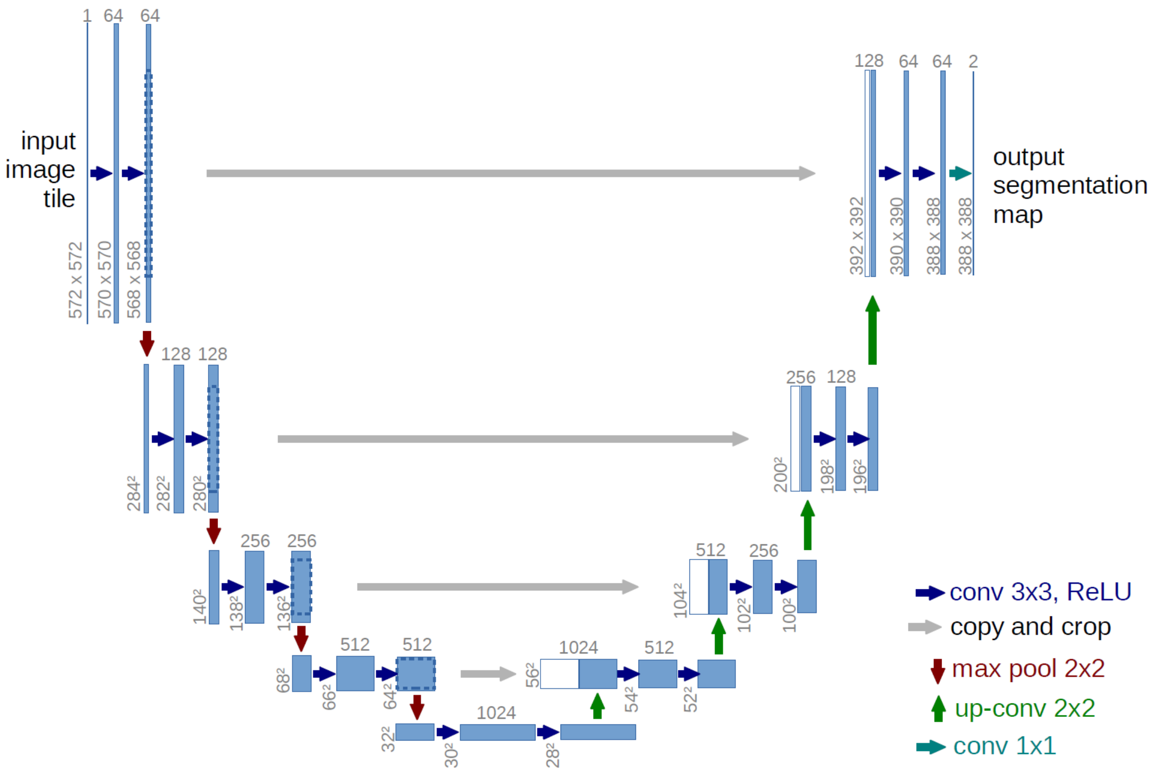

Figure 4.4: U-net Architecture

(Source:[online]Available: https://lmb.informatik.uni-freiburg.de/people/ronneber/u-net/)

The U-net is a CNN based algorithm; therefore, like any CNN, it "learns" to recognize the classes of interest using a supervised scheme by fixing the weights of convolutional filters in an iterative process [8][9]. These convolutional filters are usually organized in a network-like structure that enables the CNN to recognize spectral and spatial patterns at different scales [10][11]. The typical U-net architecture comprises of five hidden layers [12]. The U-net has two parts, an encoder and a decoder figure (4.4). In the encoder section, the input image is passed through several hidden layers. In each pass, the spatial resolution is reduced by the effect of the down sampling filters, while the "spectral" resolution is increased. On the contrary, in the decoder part, the image is passed through other hidden layers that perform the opposite process of the encoder part. Thus, in each pass, the image loses spectral resolution while it gains spatial resolution to obtain the final LULC classification. Two outputs are obtained from the U-net: the LULC classification map and the probability map.

CHAPTER 5

METHODOLOGY

The methodology of this project combines drone technology, image processing, artificial intelligence, and web development. This section details the systematic approach used in implementing the land use and land cover analysis system.

5.1 Flight Planning and Data Acquisition

- **Flight Planning**: A flight plan was designed using DJI Flight Planner around Harisiddhi to ensure systematic coverage of the study area. Later on the similar flight plan was done for the Balkhu area around college (Advanced College of Engineering and Management) premises. Flight parameters were configured based on guidelines from Tan et al. [13] which are as follows:
  - **Course Overlap Rate:** 75% (to ensure sufficient overlap between consecutive images along the flight path).
  - **Side Overlap Rate:** 65% (to ensure overlap between adjacent flight lines).
  - **Flight Altitude:** 100m
  - **Flight Speed:** 7.1m/s.
- **Image Acquisition**: High-resolution images were captured using a drone equipped with a high-quality camera. All images were geotagged with GPS data to enable georeferencing of the final output.
- **Data Organization**: The captured images were downloaded from the drone and organized into a single folder for efficient processing.

Figure 5.1: Flight Planning of Drone

5.2 Photogrammetric Data Processing

5.2.1 Image Alignment

The images were imported into Agisoft Metashape. The Align Photos tool was used to generate a sparse point cloud and estimate camera positions. Alignment settings were configured with High Accuracy and enabled Generic Preselection and Reference Preselection for optimal results. Outliers in the sparse point cloud were identified and removed to improve alignment accuracy. The Optimize Cameras tool was used to refine camera alignment parameters.

5.2.2 Dense Cloud Generation

The Build Dense Cloud tool was used to generate a dense point cloud from the aligned images. The Medium Quality setting was selected to balance processing time and output quality.

5.2.3 Digital Elevation Model (DEM) Generation

A digital elevation model (DEM) was generated using the Build DEM tool. The dense cloud was selected as the source data for DEM creation.

5.2.4 Orthomosaic Generation

The Build Orthomosaic tool was used to generate a 2D raster image (orthomosaic) from the DEM. The orthomosaic resolution was manually set to ensure high detail in the final output.

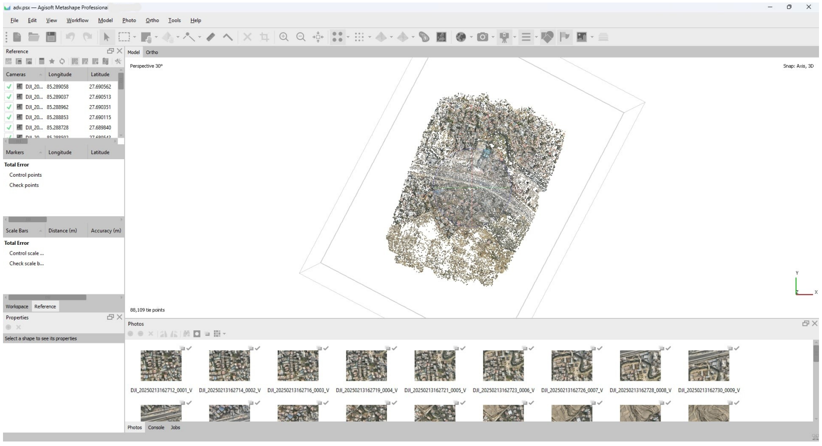

Figure 5.2: Photogrammetric Data Processing Using Agisoft Metashape

5.3 Exporting the Raster Image

The orthomosaic was exported. The output format was set to GeoTIFF for compatibility with GIS software, and the coordinate system was defined for georeferencing. The generated GeoTIFF image of Harrisiddi, Lalitpur was of size 1.4 GB having a resolution of (20334 x 26863).

Figure 5.3: GeoTIFF Image of Harisiddi, Lalitpur

5.4 Data Preprocessing and Dataset Creation

The preprocessing phase involved several crucial steps to prepare the data for AI model training:

- **Image Segmentation**: The collected GeoTIFF images were processed using Python scripts to divide them into smaller, manageable patches. Each original image was split into 9 segments to create a suitable dataset for the deep learning model.
- **Data Labeling**: Label Studio, an open-source data labeling tool, was utilized to annotate the image patches. The labeling process categorized land features into five distinct classes (building, bareland, road, water, vegetation).
- **Data Augmentation**: These large images and masks were divided into smaller patches of size 256 x 256 pixels. This was done to increase dataset size and also variability. This process helped in model's ability to generalize unseen data. We applied various transformation such as rotation, flipping to make model more robust. The augmentation was implemented using python libraries patchify, tensoflow and keras.

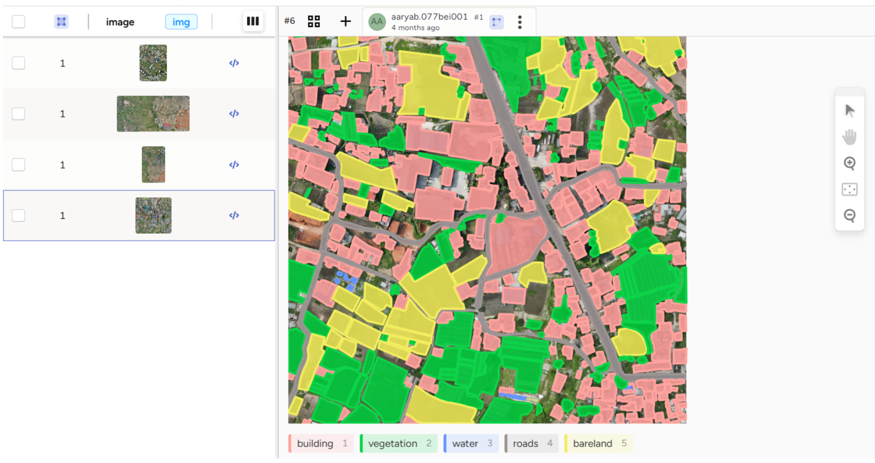

Figure 5.4: Annotation in Labelstudio

5.5 Model Development

The project implements the U-Net architecture, a specialized convolutional neural network designed for image segmentation tasks. This model was chosen for its effectiveness in handling detailed image segmentation problems and its ability to maintain spatial information throughout the processing pipeline.

5.6 Frontend Development

The Leaflet.js toolkit and its Leaflet Draw plugin were used to create the frontend of our system because of its effectiveness in handling geospatial data and producing dynamic web maps. While the Leaflet create plugin enables users to choose and create locations for land cover research, Leaflet.js renders the basic map and facilitates user interactions. Additionally, segmented land cover data is visually represented through the integration of Chart.js. Viewing and navigating aerial imagery, selecting an area using a rectangle tool, processing the selected area for prediction, and visualizing the cropped image together with its matching segmentation mask and blended output are just a few of the features offered by the online interface. Additionally, the interface lets users download and displays a legend for the many types of classified land cover.

5.7 Backend Development

The backend was developed to accept the user-selected area, process the image, and return the segmentation map with land cover classification. The backend enables seamless interaction between the frontend and our model. We used python web framework FastAPI for building API, and various python libraries such as TensorFlow, Numpy, OpenCV, etc to build fully functioning backend.

5.8 System Integration

The frontend and backend were intregated into a single outcome in web-form. After intregation, the map was displayed on the Web along with the tif file stitched in the map. The user can select a particular area to run the LULC prediction in the selected area. The selected area's coordinate is then sent to backend where the backend takes the actual tif image of the selected coordinate which then is run in the backend itself to get the result of the LULC. The result is displayed in the frontend web to the user with visible segmentation and data. The intregation step can be shown below:

- **Frontend-Backend integration**: The frontend and backend were integrated into a single web-based platform.
- **Map and TIFF Display**: After integration, the map was displayed on the web with

the overlay of the TIFF file where user can span, zoom and select specific area on the map.

- **User Selection for LULC Prediction**: Users can select a specific area on the map with the help of leaflet.js draw rectangle tool to run LULC (Land Use and Land Cover) prediction.
- **Coordinate Transmission to Backend**: The selected area's coordinates are sent to the backend for original tiff image clipping according to the coordinate sent by the frontend user.
- **TIFF Image Extraction**: The backend extracts the corresponding TIFF image of the selected area. Converts it into jpeg and is given as an input to the preloaded model in the backend.
- **LULC Processing in Backend**: The converted jpeg image is processed in the backend and given to the model to generate segmentation mask and LULC predictions. Along with these other results is also displayed on the user interface.
- **Result Display in Frontend**: The results, including segmentation and relevant data, are displayed on the user interface. The segmentation mask along with legend for 5 categories (bareland, building, roads, vegetation and water) are displayed.
- **Statistical Analysis of Land Cover Distribution**: Beyond visual outputs, the system generates quantitative insights into the selected land cover region. A statistical breakdown of the classification results is presented in a tabular format, displaying total number of pixels classified for each land category, estimated area in hectare (ha) for each class and percentage proportion of each land type in the selected area. Pie chart visualization of land use proportions generated from the results.
- **Data Export and Usability**: To facilitate further analysis, the system allows users to download processed results in multiple formats segmented images in JPG/PNG format, land classification statistics in CSV format and complete datasets in a compressed ZIP file.

5.9 System Flowchart

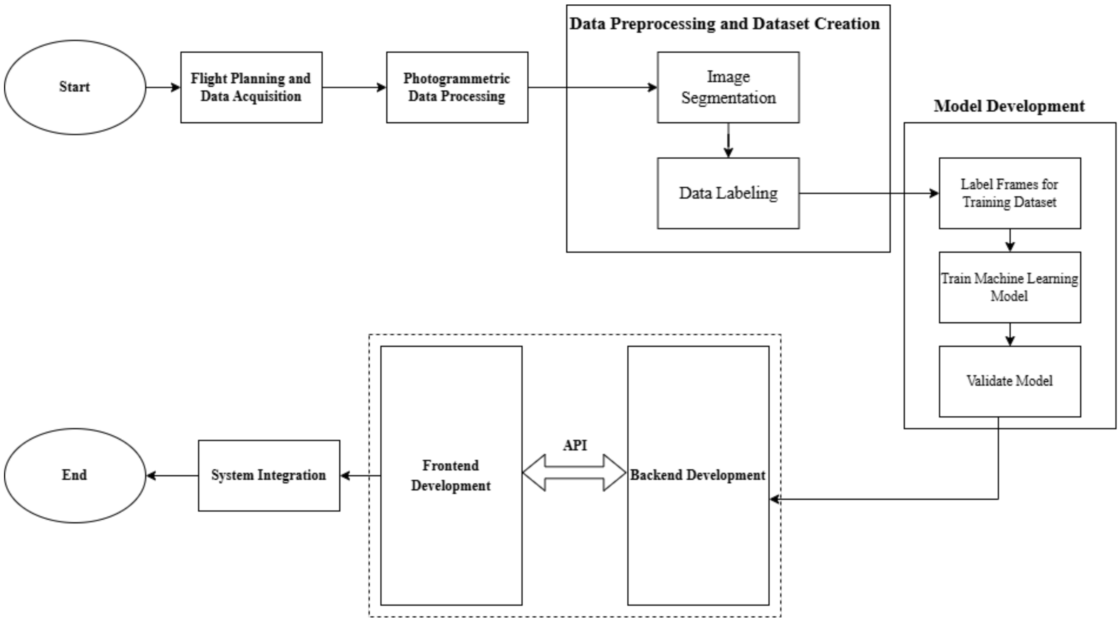

Figure 5.5: System Flowchart

CHAPTER 6

RESULT AND ANALYSIS

In this section, we present the results of our analysis and predictions regarding land use and land cover. Our study was conducted using a dataset derived from drone footage of area around Harisiddi, Lalitpur.

6.1 Ground Truth Mask Generation

The exported Common Objects in Context (COCO) format included json file which had information about the different class categories with their ID to differentiate them and their segmentation coordinated in each image with bounding-box and the information about the image that was annotated. With these information we were able to generate the ground truth mask for all images having 5 segmented categories (Bareland, Buildings, Roads, Vegetation, and Water).

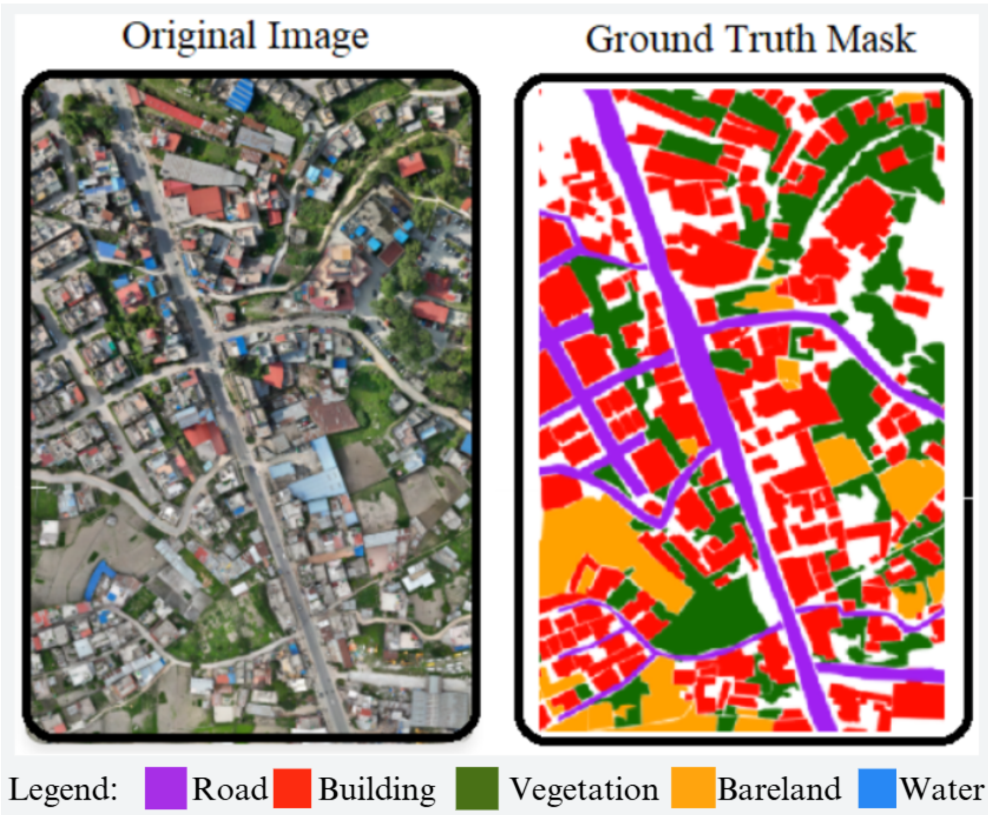

Figure 6.1: Original Image vs Ground Truth Mask

6.2 U-net Model Prediction

The proposed U-Net model is a convolutional neural network (CNN) which was trained to perform land segmentation prediction on drone-captured images, enabling precise identification of 5 different segmentation classes.

The model was able to perform pixel-wise image segmentation prediction which showed its ability to retain fine details making it ideal for high-resolution aerial imagery.

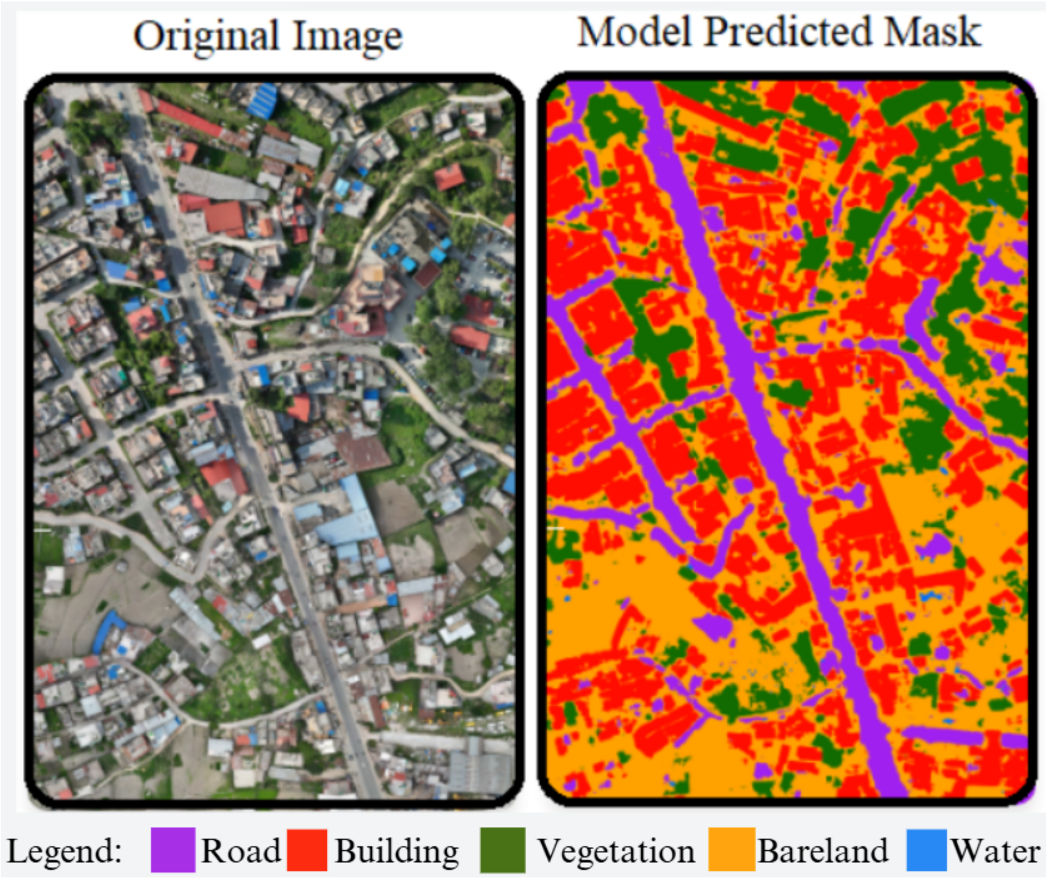

Figure 6.2: Original Image vs Model Predicted Mask

6.3 User Interface

A web interface was developed that allowed users to pan, zoom, and select regions on the GeoTIFF image overlaid on a real-world map. Leaflet's rectangle drawing tool allows the user to define the area of interest and submit them for model-based prediction and area calculations.After the model complete its prediction user can visualize their selected area and it's corresponding segmented mask predicted by the model. Users can visualize the area occupied by each segmentation class along with the interactive chart of area proportion. The users are also provided with the option to download the result which contains their selected image, its predicted mask and a csv file containing area proportion data.

6.3.1 Frontend Snapshot

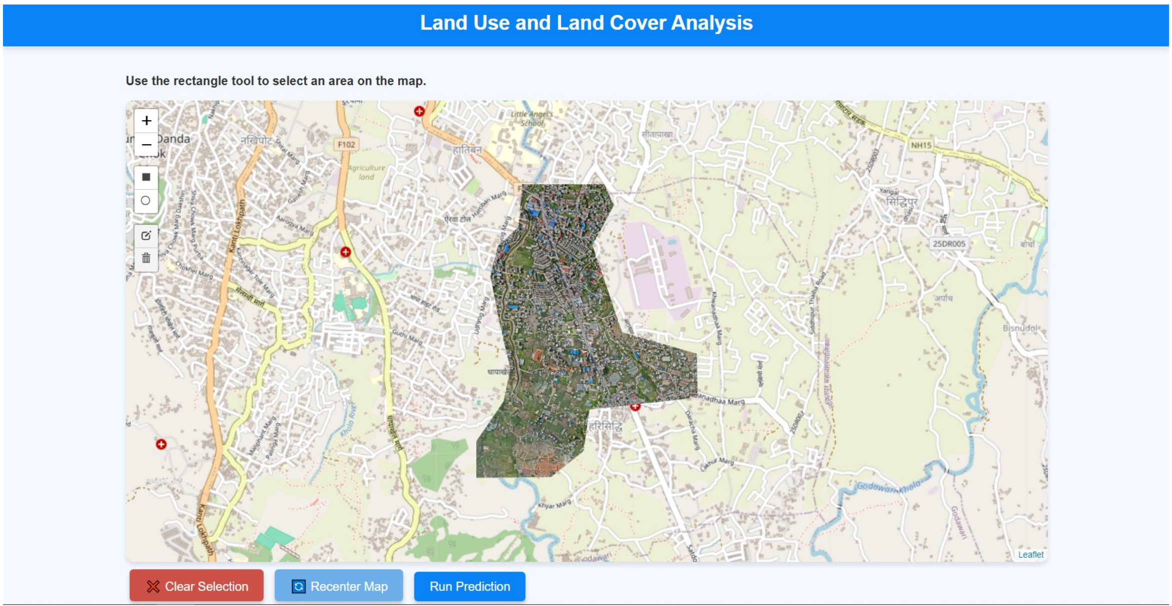

Figure 6.3: Frontend GUI

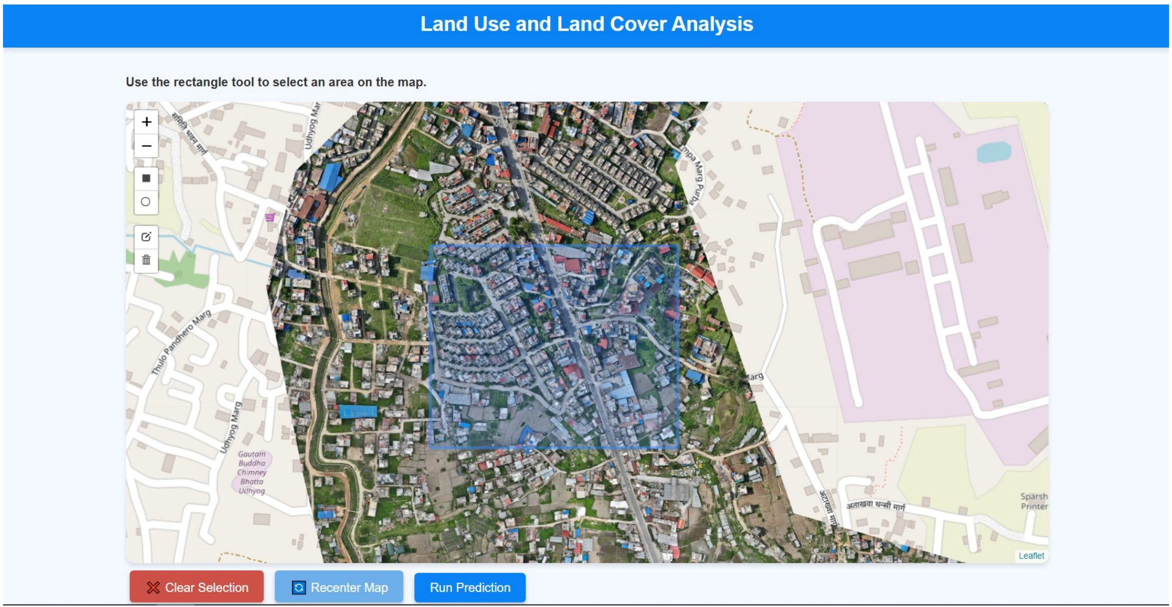

Figure 6.4: Area Selection in Frontend

6.3.2 Model Predicted Mask and Area Calculation

Category	Pixels	Area (ha)	Proportion (%)
Bareland	10130732	3.6758	39.17
Building	9559459	3.4685	36.96
Road	2945290	1.0687	11.39
Vegetation	3134316	1.1372	12.12
Water	91306	0.0331	0.35

Figure 6.5: Model Predicted Segmentation Mask Displayed on Frontend

6.4 Model Performance Evaluation

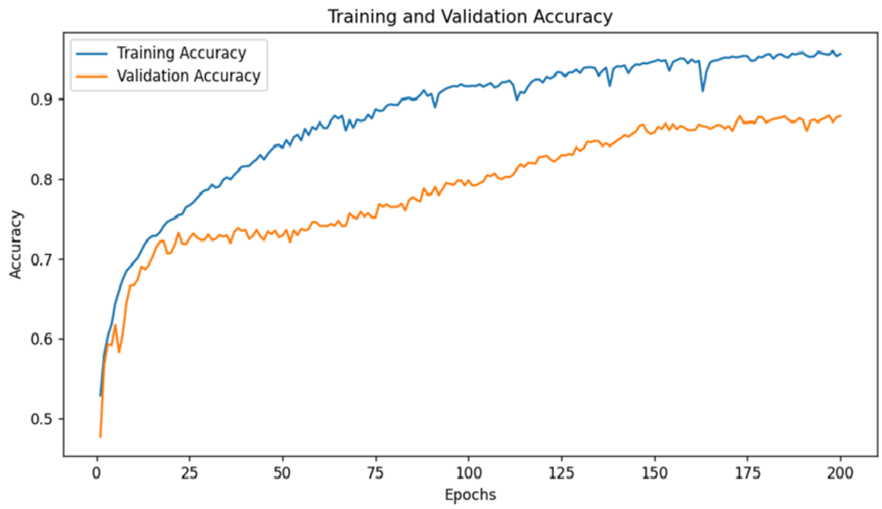

Figure 6.6: Accuracy over Epochs

The accuracy graph shows that training accuracy increases steadily, reaching approx. 95.69%, while validation accuracy stabilizes around 87.88%.

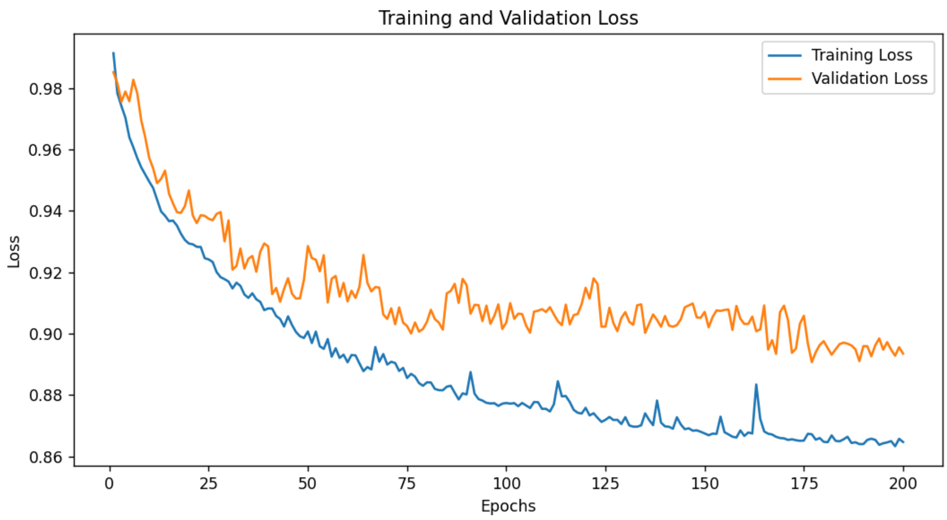

Figure 6.7: Loss over Epochs

The loss graph confirms optimization but shows fluctuations in validation loss, suggesting some instability in generalization.

6.5 Confusion Matrix

The confusion matrix demonstrates the performance of the segmentation model for various Land Use and Land Cover (LULC) categories. It specifies how many pixels were correctly and incorrectly classified for each class.

The diagonal values show how many pixels were correctly classified, while off-diagonals indicate misclassifications.

Key Observations:

- **Class 0 (Bareland)** has the highest number of correctly classified pixels (14,601,042).
- **Class 1 (Buildings)** is often misclassified as Bareland, being 723,627 pixels incorrectly labeled.
- **Class 2 (Roads)** has high misclassification rates, being confused with Bareland, as well as with Vegetation.
- **Class 3 (Vegetation)** is well classified but has 1,350,669 pixels misclassified as Bareland.
- **Class 4 (Water)** has a moderate level of classification confidence, but with confusion concerning the separating classes Bareland and Vegetation.

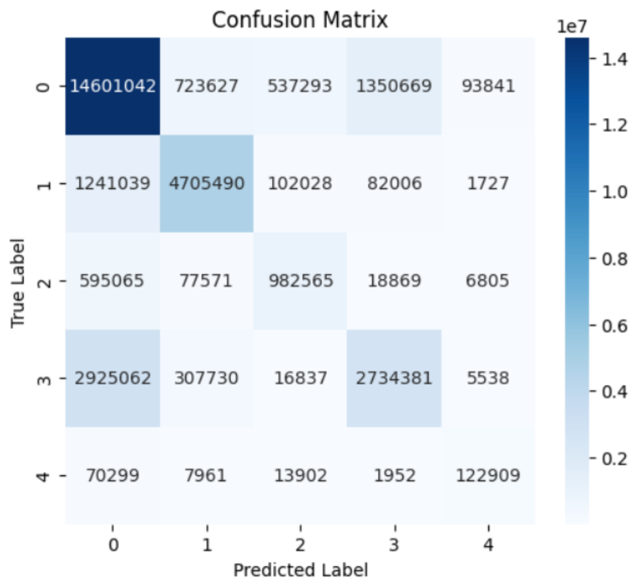

Figure 6.8: Confusion Matrix for LULC Segmentation Model

6.6 Intersection over Union (IoU) Analysis

Intersection over Union (IoU) measures how well the predicted segmentation mask aligns with the ground truth. It is calculated as:

$$IoU = \frac{|A \cap B| + 1}{|A| + |B| - |A \cap B| + 0.1} \tag{1}$$

where:

- **A** = Ground Truth Mask (`y_true`)
- **B** = Predicted Mask (`y_pred`)
- **Intersection** $(A \cap B)$ = Common pixels in both masks.
- **Union** $(A + B - A \cap B)$ = Total pixels covered by either mask.
- **(+1)** in numerator avoids zero IoU when no overlap.
- **(+0.1)** in denominator prevents instability in division.

Class	**IoU Score**
Bareland (Class 0)	0.7595
Buildings (Class 1)	0.7491
Roads (Class 2)	0.5674
Vegetation (Class 3)	0.6679
Water (Class 4)	0.4783

Table 6.1: Intersection over Union (IoU) Scores

Observations:

- **Class 0 (Bareland)** has the highest IoU score of 0.7595, indicating strong classification performance.

- **Class 1 (Buildings) and Class 3 (Vegetation)** have good IoU scores but shows some misclassifications.
- **Class 2 (Roads) and Class 4 (Water)** have the lowest IoU scores, suggesting the model struggles to differentiate these classes effectively.

6.7 Summary of Analysis

Metric	**Observation**
Best Classified Land Type	Bareland (Class 0) with IoU = 0.7595
Moderate Classification	Buildings (0.7491) and Vegetation (0.6679)
Most Misclassified	Roads (0.5674) and Water (0.4783)
Computational Performance	Optimized for real-time segmentation and web visualization

Table 6.2: Summary of Analysis

CHAPTER 7

CONCLUSION, LIMITATIONS AND FUTURE ENHANCEMENT

7.1 Conclusion

The use of Artificial Intelligence (AI) and Deep Learning (DL) for Land Use and Land Cover (LULC) mapping has transformed conventional surveying into more precise, cost-effective, and time-efficient processes. AI-based methodology, more so Convolutional Neural Network (CNN) such as U-Net, permits automated classification of geography terrain from drone-captured images.

The U-Net model, through its encoder-decoder architecture, is capable of efficiently identifying and mapping features like bare-land, buildings, vegetation, roads and water. Improves spectral and spatial resolution using convolutional filters to generate precise LULC classification maps. This automated system considerably minimizes human errors and lengthy field surveys and increases scalability and precision in large-scale geographic analysis.

Overall, AI-based LULC analysis is a huge step forward in geospatial studies, offering a cost-effective and precise alternative to traditional surveying techniques.

7.2 Limitations

- **High Resolution Imagery Dependence:** The system requires high-resolution drone images for precise Land Use and Land Cover (LULC) results.
- **Class Imbalance:** The dataset consists of imbalance classes (bareland and buildings more than water bodies) which led model for some biased model performance.
- **Temporal Limitations:** The system uses static imagery from the single point in time. The system is unable to analyze any temporal changes in land use and land cover. This is due to no available time series data regarding the land use and land cover in the study area.
- **Privacy and Regulatory Concerns:** Using drones for aerial imagery can involve pri-

vacy issues, potentially limiting operational areas or the frequency of data collection.

7.3 Future Enhancement

- **Multi-Sensor Integration:** Incorporate satellite imagery and LiDAR data to reduce reliance on drone footage. Use thermal or multispectral sensors to improve vegetation and water classification.
- **Expansion of Geographical Regions:** Expand mapping to cover additional geographic regions to cover large area for analysis.
- **Temporal Analysis:** Given the current scarcity of historical aerial imagery in Nepal, scheduling regular drone flights over designated areas to build the time series data of high-resolution images. This will help in analyzing and monitoring the land cover changes which can aid in proper urban planning.
- **Enhanced User Interactions:** Upgrade frontend with features like real-time feedback, interactive correction tools, also the ability for the users to provide annotations that can be fed back to the model for continuous improvement.
- **Cloud-Based Processing and Storage:** Use cloud technology to handle larger datasets, and provide proper scalability platform.

REFERENCES

[1] B. Kalantar, S. B. Mansor, M. I. Sameen, B. Pradhan, and H. Z. Shafri, "Drone-based land-cover mapping using a fuzzy unordered rule induction algorithm integrated into object-based image analysis," *International journal of remote sensing*, vol. 38, no. 8-10, pp. 2535–2556, 2017.

[2] Y. O. Ouma, A. Keitsile, B. Nkwae, P. Odirile, D. Moalafhi, and J. Qi, "Urban land-use classification using machine learning classifiers: Comparative evaluation and post-classification multi-feature fusion approach," *European Journal of Remote Sensing*, vol. 56, no. 1, p. 2 173 659, 2023.

[3] E. A. Alshari, M. B. Abdulkareem, and B. W. Gawali, "Classification of land use/land cover using artificial intelligence (ann-rf)," *Frontiers in Artificial Intelligence*, vol. 5, p. 964 279, 2023.

[4] A. Alem and S. Kumar, "Deep learning methods for land cover and land use classification in remote sensing: A review," in *2020 8th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions)(ICRITO)*, IEEE, 2020, pp. 903–908.

[5] M. Carranza-García, J. García-Gutiérrez, and J. C. Riquelme, "A framework for evaluating land use and land cover classification using convolutional neural networks," *Remote Sensing*, vol. 11, no. 3, p. 274, 2019.

[6] J. Jagannathan and C. Divya, "Deep learning for the prediction and classification of land use and land cover changes using deep convolutional neural network," *Ecological Informatics*, vol. 65, p. 101 412, 2021.

[7] "Dji mavic 3 enterprise series_industrial-grade surveying and inspection drone_dji enterprise applications." (2023), [Online]. Available: `https://enterprise.dji.com/mavic-3-enterprise`.

[8] N. Buduma, N. Buduma, and J. Papa, *Fundamentals of Deep Learning*. Boston, MA, USA: O'Reilly Media, Inc., 2022, p. 283.

[9] F. Chollet, J. Allaire, and P. Team, *Deep Learning with R*. Shelter Island, NY, USA: Manning Publications Co., 2018, p. 335.

[10] T. Kattenborn, J. Leitloff, F. Schiefer, and S. Hinz, "Review on convolutional neural networks (cnn) in vegetation remote sensing," *ISPRS journal of photogrammetry and remote sensing*, vol. 173, pp. 24–49, 2021.

[11] X. Yuan, J. Shi, and L. Gu, "A review of deep learning methods for semantic segmentation of remote sensing imagery," *Expert Systems with Applications*, vol. 169, p. 114 417, 2021.

[12] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III*, ser. Lecture Notes in Computer Science, vol. 9351, Cham, Switzerland: Springer, 2015, pp. 234–241.

[13] C. Tan, Z. Chen, A. Liao, and X. Zeng, "Accuracy assessment of mavic 3 industrial uav based on dji terra and pix4dmapper," in *Fifth International Conference on Geology, Mapping, and Remote Sensing (ICGMRS 2024)*, SPIE, vol. 13223, 2024, pp. 498–504.

APPENDIX : ACEM Drone Orthomosaic

Drone Orthomosaic of Advanced college area, Balkhu, Kathmandu